AI PRODUCT MANAGEMENT CASE STUDY · WEEK ONE DONE, WEEK TWO NEXT · OCTOBER 2026

Clarivi

A plain-words morning answer from my own Apple Watch data

For Apple Watch users who train and sleep with the watch on. It says what last night's readings mean against your own normal, why the day's status was set, and one thing to do today. I built it and tested it on myself, and I report what missed next to what worked.

The problem

Apple Watch and the Health app collect an unusually rich stream of physiological and behavioural data (heart rate, heart rate variability, sleep, workouts, weight and gait), but mostly show each metric as an isolated daily card, with little sense of trend. There is no way to see how a metric is drifting over time, how it relates to other metrics, or whether today's reading is actually unusual for this person. This project builds a tool that takes a person's own Apple Watch data each morning and turns it into a personalised, longitudinal picture of their own recovery, sleep and training patterns.

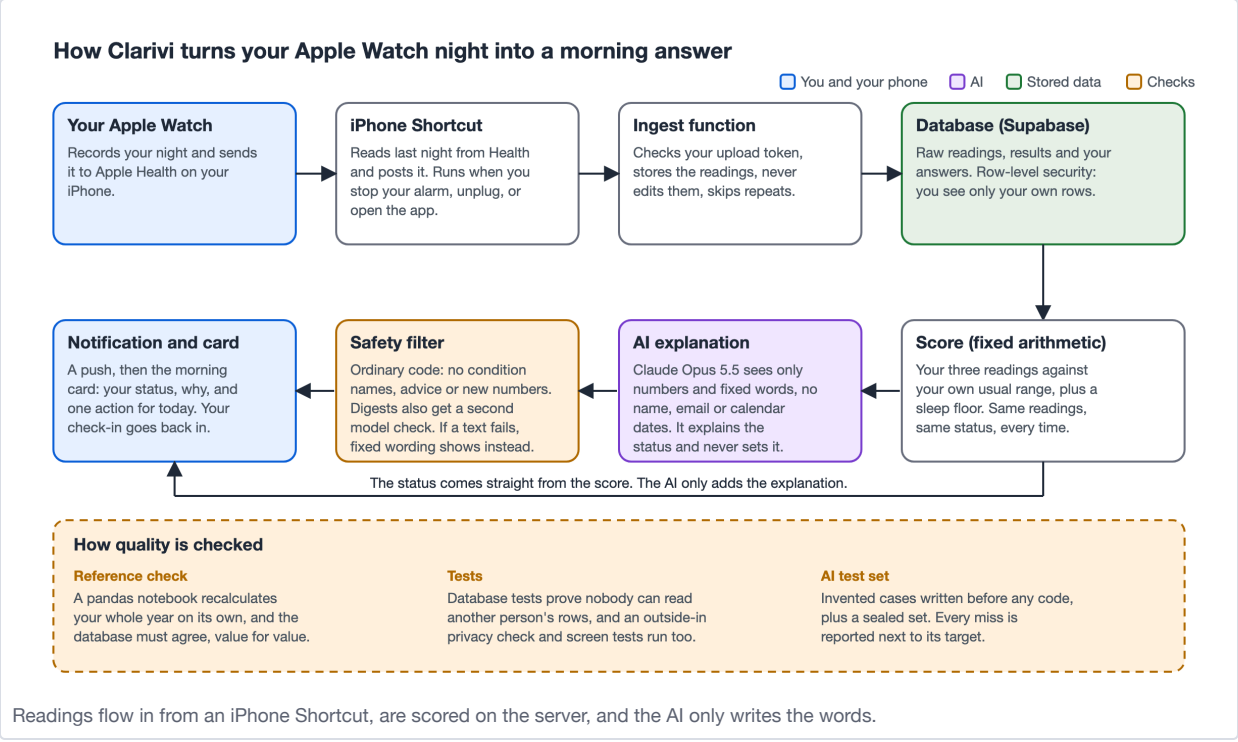
- **Who it is for:** Apple Watch users who train consistently (strength training, general fitness) and already track workouts and sleep, but want to understand the patterns behind their numbers rather than just seeing isolated daily readings.
- **What problem it solves:** Apple Health has no way to detect drift against a person's own baseline, surface relationships between metrics (for example sleep and next-day recovery), or tell a normal day from an unusual one.
- **Why it matters:** without that visibility, early signs of under-recovery, overtraining or illness go unnoticed until performance or health has already suffered, and users cannot tell whether a change they made (more sleep, a deload week) actually worked.
- **What the user should be able to do:** wake up and get a plain-words answer from their own data: whether last night, set against their own normal, is meaningfully off, why, and one action for the day, with the past weeks of each reading one tap away.
- **The outcome:** earlier, evidence-based awareness of their own recovery and readiness, so training and lifestyle adjustments are driven by their own data rather than generic guidance.

261 nights
My year's nights, usual ranges and daily status agree with an independent check: 0 values differ. Same author, so independent code, not an independent person.

38 of 40
Invented cases passed the AI's safety filter on one live attempt. The filter was tuned on these cases.

4 of 7
Days with a sync in week one, against a target of 6: missed. Only 2 of 7 had a status before noon.

2 of 13
Disrupted mornings the score flagged, against a target of 2 in 3: missed. One person, thin evidence.



Readings flow in from an iPhone Shortcut, are scored on the server, and the AI only writes the words.

What I built

- **The card:** Ready, Ease off or Rest in a few plain sentences, one action for the day, and a "Why today" screen that shows how it was decided.
- **The score:** fixed arithmetic in SQL against my own last 42 nights, checked by separate pandas code. A night with sleep far below my usual cannot be Ready.
- **The AI:** Claude Opus 5.5 writes the wording and never sets the status. It sees an allow-list of results only, and its text goes through a filter, a retry and a fixed fallback.
- **The sync:** an iPhone Shortcut on three triggers (alarm stopped, charger unplugged, app opened), with an 11:30 reminder if the status is missing.
- **Private by default:** row-level security on every table, no third-party scripts, three separate consent boxes.

Key decisions

- **A fixed score, not a learned model:** every status is explainable and checkable. The cost: it cannot adapt to me.
- **The AI explains, never decides:** a wrong sentence cannot change what the card says. The cost: a short wait, and a US AI provider that needs its own consent.
- **A smaller test, set before any result:** four people for 28 days became one person for a week. The cost: it cannot speak for anyone else.
- **Test before building:** I tried three planned analysis modules on my own year first. Two found nothing the data could support (0 of 12 each), so I built none.

How I measured quality

Ten targets, set on 29 September 2026, before any result. Week one, 19 to 25 October:

Measure	Target	Result
Sync	6 of 7 days	**4 of 7**
Correctness	Match the reference	**0 differ**
Isolation	No tolerance	**Held**
Signal, my year	2 in 3	**2 of 13**
Quiet by default	About 1 in 7	**52 of 231 days**
Engagement	5 of 7 days	**3 of 7**

What I learned

- The sync miss was partly the Watch: its sleep can reach Health hours after waking. A fixed wait failed on its first real morning, so I dropped it and made the reminder cover an unfinished night.
- A Ready on a very short night led to a sleep floor. It made the score louder than my "quiet" target (52 of 231 days, against 29 of 231 before it), and I report that as a miss.
- Counting the sync three ways showed that "4 of 7" counted any post, so week two counts the status shown before noon.

> **Limits:** one person, one week so far. The same author built the system and counted the results. Week two is not included yet.

In the repository: README, docs/evaluation.md, Project Memory (specs and plan), the architecture diagram and the checks.

Skills: AI product management (specs, phased plan, success targets set first, evaluation, iteration from data), responsible AI and data protection.